Marc Berthelette

Senior DevOps Engineer — Automation, CI/CD and Guardrails for AI-Assisted Engineering

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Bilingual: French / English · Incorporated in Canada · Available immediately

Professional Summary

Senior DevOps engineer with nearly 18 years across infrastructure, development and operations. Introduced AI-assisted engineering practices at Énergir with hard scope gates and human-validation checkpoints before any AI-generated code reaches production. Built shared CI/CD for 12 teams and 70+ applications and a Python/FastAPI validation service at Desjardins that reports to Dynatrace. Works from Python and Azure; Terraform and Kubernetes experience is as a user of platforms other teams operate.

Technical Skills

Automation & Programming: Python, FastAPI, Bash, Java, C++, SQL
CI/CD: Jenkins, GitHub Actions, Azure DevOps, SonarQube quality gates, JFrog Artifactory
Cloud & Infrastructure: Azure, Terraform/Ansible workflows (user of another team's modules), Linux, AWS migration environment
Containers & Observability: Kubernetes application deployments, ArgoCD, Docker, Dynatrace, Splunk, ELK

Work Experience

Énergir — Senior DevOps Engineer (Contract)

Montréal, QC | September 2021 – September 2026
Five-year consulting engagement, renewed four times.

- Introduced AI-assisted engineering practices with hard scope gates and human-validation checkpoints before any AI-generated code reaches production.
- Initiated shared Jenkins pipelines adopted by all teams already deploying through Jenkins, in an environment of about 50 developers, 12 teams and 70+ applications; achieved approximately 80–90% JaCoCo coverage of shared pipeline code with per-commit unit tests and built sample applications for integration testing.
- Enforced shared stages and mandatory SonarQube checks in the shared pipelines so developers could not simply skip them.
- Separated application-code errors from platform-pipeline errors in build reporting so teams could see whose fix a failure was.
- Replaced legacy ANT deployment scripts with the official Salesforce CLI and built validation that deployed each candidate merge into a dedicated sandbox and ran the full test suite.
- Migrated production Salesforce logs to direct Dynatrace ingestion; expanded error parsing and dashboards to help the team investigate previously hidden issues.
- Replaced a 15-step MuleSoft checklist with version-based deployments of QA-tested artifacts, moving release preparation ahead of production windows and simplifying rollback.
- Ran recurring enablement sessions (about six per year) with each application team after pipeline delivery; onboarded new DevOps engineers and developers and was the primary developer contact when the delivery ecosystem broke.

Key technologies: Jenkins, SonarQube, Azure, Python, JaCoCo, Salesforce CLI, Oracle, Node.js, MuleSoft, Dynatrace

Desjardins — Senior DevOps Engineer (Contract)

Montréal, QC (Remote) | October 2025 – March 2026
Concurrent with the Énergir engagement.

- Built a Python/FastAPI workflow that provisioned Artifactory, applied configuration from a selected Git reference, and tested uncached artifact retrieval across approximately 250 repositories.
- Deployed the suite through ArgoCD/GitOps for scheduled execution, publishing configuration-version and artifact-level results to Dynatrace to expose configuration and upstream connectivity failures.

- Made most validation logic tool-agnostic and the artifact test set configurable so platform administrators could add their own cases.
- Implemented artifact retention policies and Splunk alerts for configuration changes made outside CI/CD.

Key technologies: Python, FastAPI, Artifactory, Kubernetes, ArgoCD, Dynatrace, Splunk

Tink — DevOps Engineer

Montréal, QC | January 2018 – August 2021

- Replaced reactive patching across 600+ servers with staged Spacewalk schedules, environment/role groups, A/B frontend and middleware batches, and dependency-aware reboots.
- Built fleet inventory across Unix and Windows servers to identify vulnerable applications and upcoming end-of-life upgrades; added TLS scoring and environment version tracking for Ops, product owners, and QA.
- Automated PRTG-triggered response to Elasticsearch CPU saturation: attempted a process restart after two minutes and alerted on-call at five minutes if saturation persisted; supported a 24/7 rotation.
- Built centralized ELK logging and managed fleet configuration with Puppet.
- Built an AWS migration target using console provisioning (EC2, S3, Route 53, VPC, caching and database services); the client cancelled before production cutover.

Key technologies: Linux, Jenkins, Puppet, Spacewalk, PRTG, ELK, AWS, Rocket.Chat

SOVO Technologies — Lead Technician → Software Engineer

Montréal, QC | November 2008 – December 2017

- Led and performed most redundancy work across HAProxy, authentication, speech-recognition servers and file services with a second technician; supported live-captioning requirements of three minutes maximum downtime per broadcast and eight-second caption latency.
- Owned IT planning and execution for two office relocations involving approximately 15–20 workstations, preparing networks and cabling and coordinating phased moves to minimize disruption.
- Developed production scheduling, speech-recognition evaluation, and multilingual dictionary software; served as the final quality gate before production releases.

Key technologies: HAProxy, Active Directory, VMware vSphere, Linux, Windows Server, C++, Java, SQL

Education

Bachelor of Automated Manufacturing Engineering — École de Technologie Supérieure (ÉTS), Montréal | 2016

DCS in Computer Science Technology — Cégep André-Laurendeau, Montréal | 2005